Artificial Intelligence

Module 01

Foundations of Artificial Intelligence

Learning Objectives

By the end of this session, participants will be able to:

1. Correctly explain what modern AI is **and what it is not**.
2. Identify **core capabilities vs. structural limitations** of AI systems.
3. Recognize **common failure modes** (hallucinations, bias, privacy risks).
4. Apply **responsible-use principles** to real workplace tasks.
5. Reflect on **when AI should vs. should not** be used in their own role.

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Why AI Matters Now?

In 2026, AI has moved from a "novelty" to a "utility".

1. **Productivity (Automating Cognitive Labor)**: AI acts as a "multiplier" for human effort. It doesn't replace the worker; it replaces the **friction** in their workflow. Think of AI as "Industrial Revolution 2.0." Just as steam engines replaced physical muscle, AI is replacing "cognitive muscle"—repetitive tasks like summarizing, drafting, and organizing.
2. **Speed (Real Time Everything):** Humans process data linearly (one page at a time). AI processes information massively in parallel. AI can synthesize a 500-page legal document or a year of global market data in seconds, providing "High-Frequency Insights" that allow for instant decision-making.
3. **Scale (Navigating "Hyper Complexity")**: Human brains are generally limited to tracking 4-7 variables at once. AI can track thousands of dimensions simultaneously. In logistics and medicine, AI evaluates millions of combinations (e.g., drug interactions or delivery routes) that are statistically impossible for a human team to calculate.

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AI Won't Replace Humans - But Humans With AI Will Replace Humans Without AI.

Karim R. Lakhani, Harvard Business School Professor

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AI Applications – Passive Use

This slide explores the **invisible AI infrastructure** that powers your daily digital life. These are "passive interactions"—meaning the AI works in the background to simplify your day without you ever having to press a "Start AI" button.

1. **Optical Character Recognition (OCR)**: Uses Machine Learning (ML) to recognize shapes as letters and numbers, then converts them into digital data. Modern AI enhanced OCR (Intelligent Character Recognition) now achieves over 99% accuracy on complex handwritten document.
2. **Recommendation Engines:** Collaborative filtering and deep learning analyze your "dwell time" (how long you look at a post), likes, and shares to predict your future interests. Approximately 90% of content users watch on TikTok is recommended by the AI rather than from the accounts they actually follow.
3. **Ride Sharing (Ride/Feres)**: AI balances supply (drivers) and demand (riders) while calculating the fastest route through live traffic.
4. **Email Spam Filters**: Natural Language Processing (NLP) and "Bayesian filtering" analyze the intent, sender reputation, and metadata of every incoming email.

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AI Applications - Active Use

If passive AI is the invisible infrastructure, active AI is your collaborative partner. These interactions are intentional, you provide a prompt, a file, or a goal, and the AI generates a specific output.

1. **Large Language Models (LLMs)**: ChatGPT (OpenAI), Claude (Anthropic), Gemini (Google). Works on the transformer architecture. Gartner projects that by the end of 2026, over **100 million people** will use generative AI as a primary "co-worker" for daily tasks.
2. **AI Visual Creation (Diffusion Models)**: Midjourney, Nano Banana, Sora/Veo. These models are trained by taking an image and slowly adding "static" (noise) until its unrecognizable. To generate an image, the AI starts with a block of pure noise – and guided by your text prompt – mathematically "denoises" it, pulling a clear image out of the static.
3. **AI Agents & Task Orchestrators**: Zapier, CrewAI, Microsoft Copilot Agents. Unlike a chatbot that just talks, an Agent uses a "Plan-Act-Observe" loop. It breaks a complex goal into sub-tasks.
4. **Coding Copilots**: GitHub Copilot, Cursor. These models are trained on billions of lines of open-source code.

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What is Artificial Intelligence?

Artificial intelligence (AI) is a set of technologies that empowers computers to learn, reason, and perform a variety of advanced tasks in ways that require human intelligence, such as understanding languages, analyzing data, and even providing helpful suggestions.

It encompasses many different disciplines, including computer science, data analytics and statistics, hardware and software engineering, linguistics, neuroscience, and even philosophy and psychology.

Artificial intelligence techniques, though diverse, all fundamentally rely on data, algorithms, and computational power. AI systems learn and improve through exposure to vast amounts of data, identifying patterns and relationships that humans might miss. This data serves as the training material, the quality and quantity of which are crucial for the AI's performance.

Source: https://cloud.google.com/learn/what-is-artificial-intelligence

What is Artificial Intelligence?

Let's break AI down in a way that everyone can understand.

- **What it is**: AI is a field of computer science focused on creating smart machines that can perform tasks that typically require human intelligence, like learning, reasoning, and problem-solving.
- **How it works**: AI systems learn from vast amounts of data, identifying patterns to make predictions or decisions without being explicitly programmed for every scenario. Think of it as teaching a computer by showing it a million examples instead of writing a million rules.
- **Where you see it**: You're using AI every day in navigation apps like Google Maps, personalized recommendations on shopping sites, spam filters in your email, and virtual assistants like Gemini Live.
- **Why it matters**: AI helps us solve some of the world's toughest challenges—from accelerating medical research to creating more efficient supply chains and tackling climate change.

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Source: https://cloud.google.com/learn/what-is-artificial-intelligence

Artificial Intelligence Technologies

AI isn't a single technology but a broad field encompassing several key areas:

- **Machine Learning (ML):** This is a type of AI where systems learn from data to identify patterns and make predictions or decisions without direct programming. Imagine teaching a computer to recognize a bird by showing it thousands of bird pictures; it learns what a bird looks like on its own.
- **Deep Learning (DL):** A subfield of ML, deep learning uses artificial neural networks with many layers (hence "deep") to learn from data. These networks are inspired by the structure of the human brain and are particularly good at complex tasks like image and speech recognition.
- **Natural Language Processing (NLP):** NLP enables computers to understand, interpret, and generate human language. This is what powers voice assistants like Siri and Alexa, translation services, and chatbots.
- **Computer Vision:** This area allows computers to "see" and interpret visual information from the world, such as images and videos. It's used in everything from facial recognition to self-driving cars.

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Story of Artificial Intelligence

1. The earliest AI systems (1950s – 1970s) were overwhelmingly rule-based (Symbolic AI). At this time artificial intelligence was thought to come from **reasoning over symbols**, not learning from data. It was not the only approach; it simply was the dominant one.
2. In parallel, neural networks existed very clearly (McCulloh-Pitts Neuron, 1943 and Rosenblatt's Perceptron, 1957), but they were computationally weak, poorly understood, and temporarily discredited after Minsky & Papert (1969).
3. Because of this, Symbolic/Rule based AI dominated for decades, while learning based methods went into an "AI winter".
4. Rule based AI eventually failed because it collapses under real-world complexity.
    1. Rules don't scale (combinatorial explosion).
    2. Knowledge is brittle and hard to maintain.
    3. No learning $\longrightarrow$ No adaptation.
    4. Hidden assumptions everywhere.

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Story of Artificial Intelligence

Symbolic AI failed because of the below natural principle:

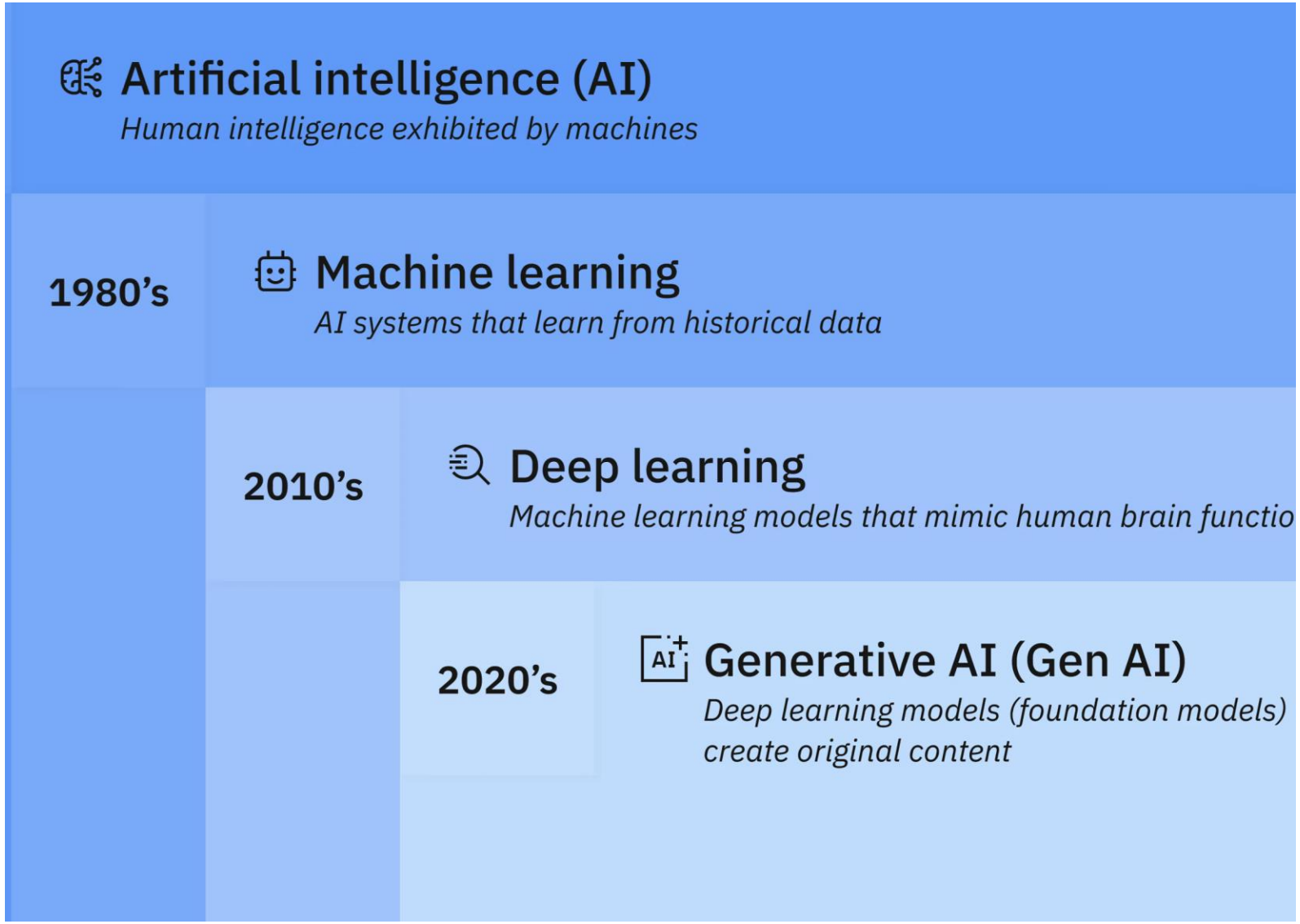
Intelligence ≠ explicit rules
Intelligence ≈ pattern discovery + generalization

Symbolic AI was replaced by Statistical ML (Bayesian models, decision trees) in the 1980s-90s, and then by data-driven ML at scale (2000s), and from 2010 onwards, deep learning, transformers, and foundation models have been the more revolutionary approaches to Artificial Intelligence.

Clean mental model:

1. Rule-based AI - "We tell the machines how to think."
2. Machine Learning AI - "The machine learns how to think."

Evolution of Modern AI Technologies



Rule Based Systems

Rule-based systems, a foundational technology in artificial intelligence (AI), have long been instrumental in decision-making and problem-solving across various domains. These ***deterministic*** *systems operate on a set of predefined rules and logic (if-then) to make decisions, perform tasks, or derive conclusions. Despite the rise of more advanced AI methodologies, such as machine learning and neural networks, rule-based systems remain crucial due to their transparency, ease of use, and interpretability.*

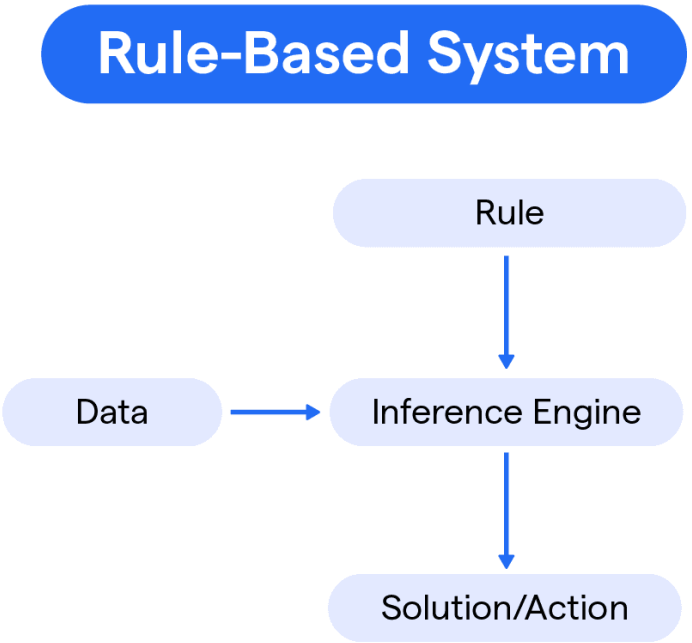
Some applications include:

1. **Fraud Detection:** Credit card companies use rules like "If a transaction happens in 2 countries within 5 minutes, then flag it as suspicious."
2. **Access Control & Security:** Firewall rules, spam filters, and authentication systems often use deterministic logic: "If password has < 8 characters, then reject."

Rule Based Systems

Comprising of a knowledge base and an inference engine, these systems provide high transparency, consistency, and reliability for automating structured, repetitive tasks. Key aspects of a rule-based system are:

1. **Structure**: They consist of a knowledge base, working memory, and an inference engine.
2. **Mechanism**: Operates on deductive logic; If a specific condition (IF) is met, the system executes a corresponding action (THEN).
3. **Deterministic**: Outcomes are predictable and consistent because the system only follows programmed rules, unlike Machine Learning, which infers patterns from data.
4. **Applications**: Commonly used in expert systems, chatbots, decision support systems, and for regulatory compliance.



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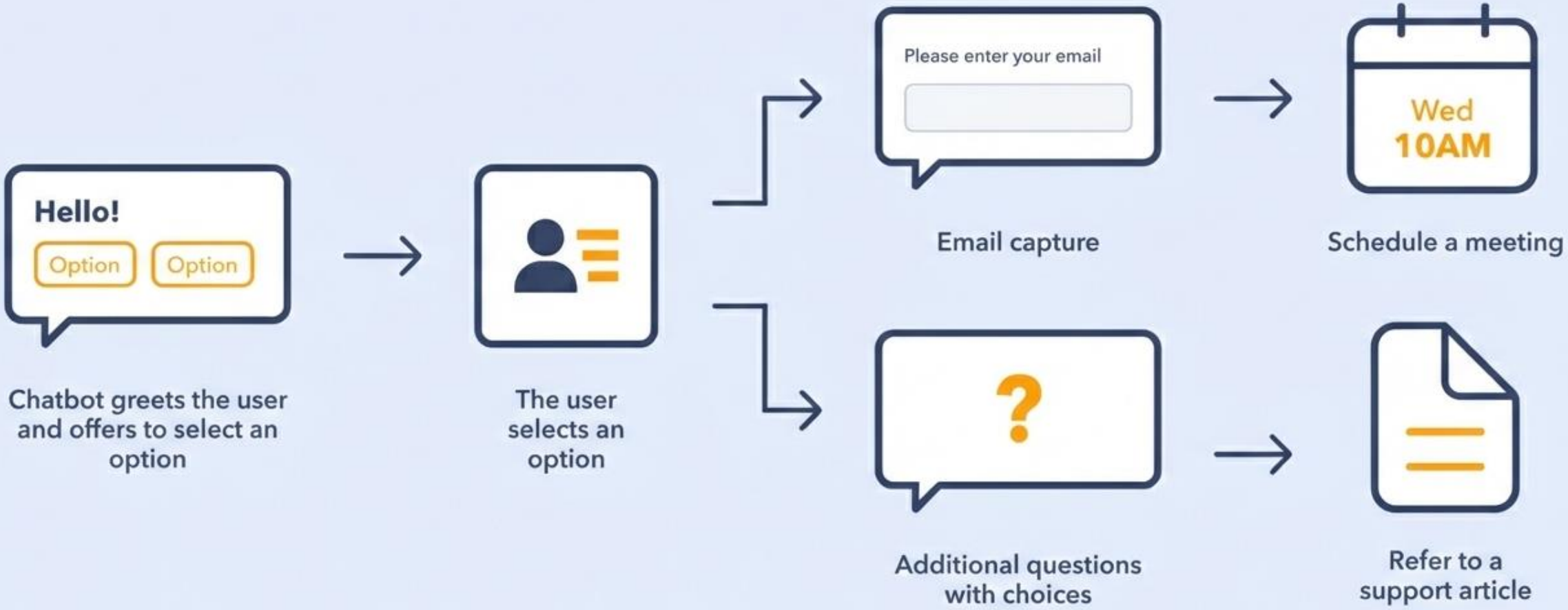
How A Rule-Based Chatbot Works



Rule Based Systems: Pros and Cons

Advantages	Disadvantages
Transparency: Decisions are easy to trace and explain, known as "white-box" AI.	Scalability: Difficult to manage or scale as the number of rules increases.
Consistency: Eliminates human error and bias in repetitive tasks.	No Learning: Cannot adapt, learn, or improve on their own.
No Training Data: Does not require large datasets to operate.	Rigidity: Fails in complex, dynamic, or unforeseen situation.

While modern AI often favors machine learning, rule-based systems remain highly effective for, and superior to, scenarios requiring strict, documented logical compliance.

Search vs. GenAI

Feature	Search Engine (e.g., Google)	Generative AI (e.g., ChatGPT)
Primary Goal	**Finding:** To locate specific, existing information or sources.	**Creating:** To synthesize, summarize, or generate new content.
Core Mechanism	**Indexing & Ranking:** Scans a massive database (the web) to find the best match.	**Next-Token Prediction:** Calculates the statistical probability of the next word[3].
Nature of Output	**Deterministic:** For the same query and index, the sources remain consistent.	**Probabilistic:** Can produce different (and "creative") results for the same prompt.
Relationship to Truth	**Source-Based:** Success is measured by the accuracy and authority of the link.	**Pattern-Based:** Success is measured by how "plausible" and helpful the response sounds[4].
Best Used For	Facts, news, navigating to sites, and verifying original sources.	Drafting, brainstorming, coding, and explaining complex concepts[5].
The "Failure Mode"	**Broken Links:** Information might be outdated or missing.	**Hallucinations:** The system confidently predicts a pattern that isn't true[6].

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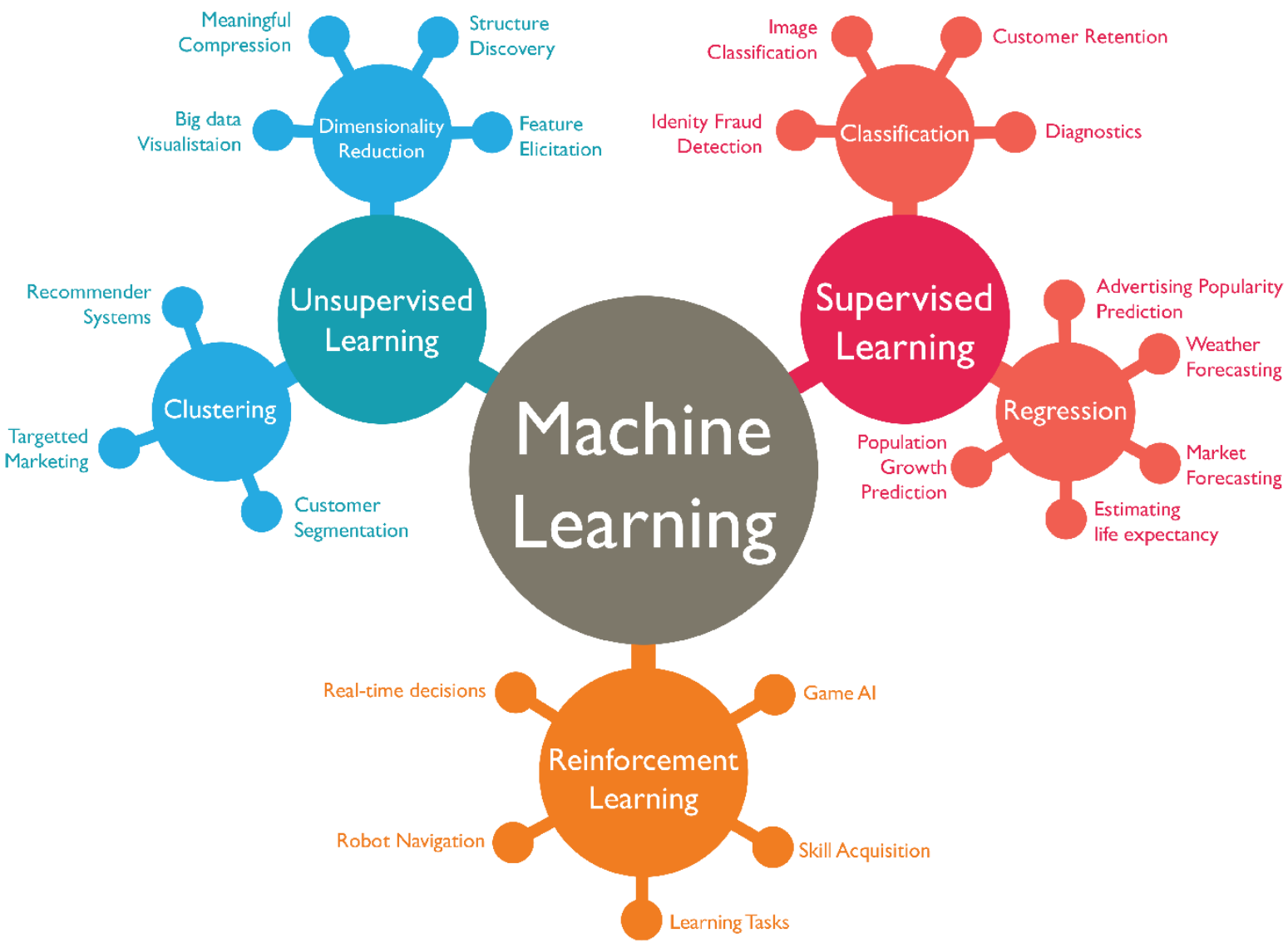
Search vs. GenAI

"Probability vs. Truth" concept:

- **"Search is a Librarian; GenAI is a Storyteller":** A search engine walks into a library to find the exact book you asked for; in contrast Generative AI has read every book in the library and is now trying to tell you a *new* story based on what it remembers.
- **The "Not a Database" Rule:** Because GenAI is not looking things up in a table, it doesn't "know" facts. It only knows that in its training data, certain words usually appear together.
- **The Resulting Risk:** This explains why you must verify GenAI but can usually trust a top search result to lead to a real website.

Machine Learning



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Supervised Learning

Supervised learning is a type of machine learning where the model is trained on a **labeled dataset**. This means every piece of input data (x) is paired with the correct output label (y). The goal is for the model to learn a rule (a mathematical function) that maps inputs to outputs so accurately that it can predict the label for new, unseen data.

Two Main Tasks:

- **Classification:** Predicting a category (e.g., Is this transaction "Fraud" or "Normal"?).
- **Regression:** Predicting a continuous number (e.g., Predicting the price of a house in Addis Ababa based on its square footage).

Feedback Loop: During training, the model makes a guess, compares it to the "ground truth" (the correct label), and adjusts its internal settings to reduce the error.

Overfitting: A common risk where the model "memorizes" the specific training examples rather than learning the general pattern, making it fail on real-world data.

Unsupervised Learning

Unsupervised learning is a type of machine learning where the computer is given data without any specific labels or "correct answers."

Imagine giving a child a large pile of different seeds. You don’t tell them which is "maize" or which is "teff." Instead, the child looks at the pile and naturally begins to group them based on their shape, size, and color. The computer does the same thing: it looks for **hidden patterns** or structures within the data on its own.

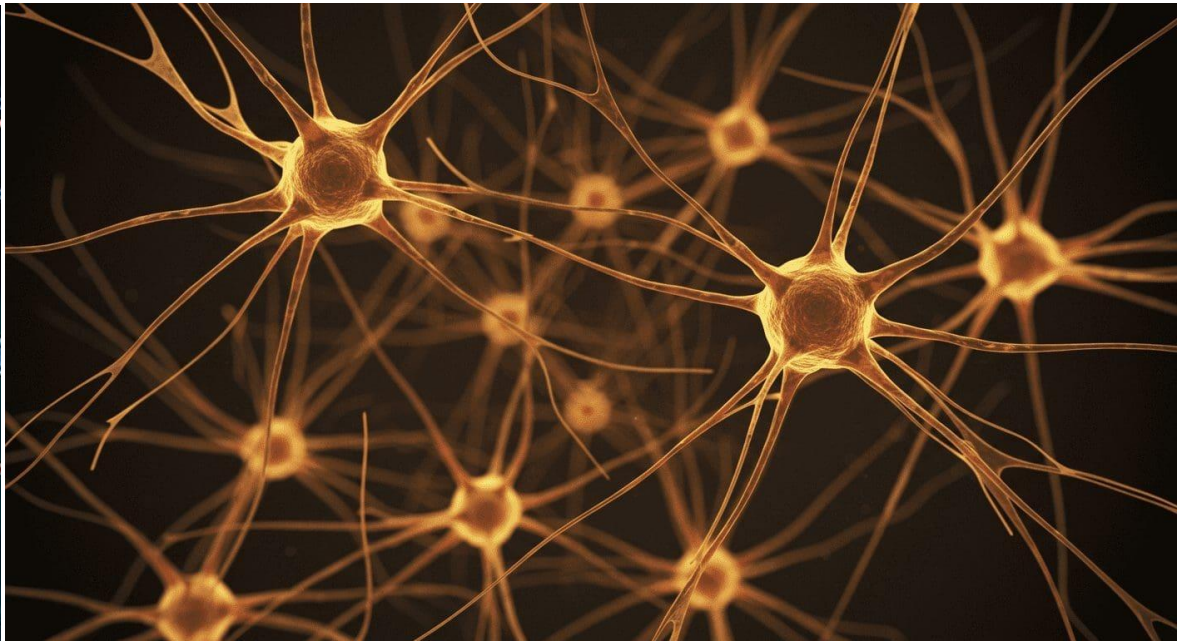
Example: Identifying "Customer Types" for Ethio Telecom or Safaricom

Telecommunications companies have millions of users. They use unsupervised learning to "cluster" customers based on behavior:

- **Group A:** People who mostly use data late at night (perhaps students or night-shift workers).
- **Group B:** People who only make short calls during business hours (likely merchants or professionals).
- **Group C:** People who rarely call but receive many international transfers (families with relatives abroad).
- **Result:** The company discovers these distinct groups and can create specific packages (like a "Student Night Data" plan) that fit each group's lifestyle.

Quiz

Can you recognize what this is?

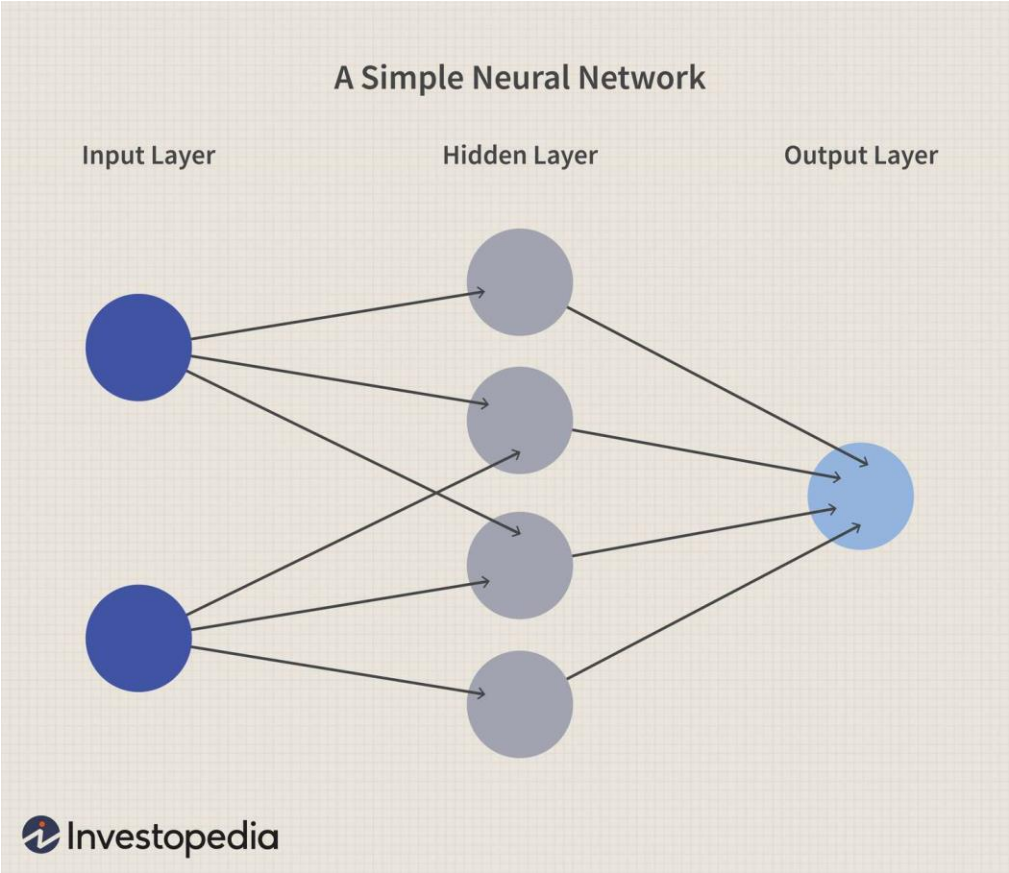


Neural Networks

A neural network is a computational model inspired by the human brain that learns to recognize patterns in data. It is a system of **interconnected nodes** (artificial neurons) organized into layers that map an input to an output through mathematical transformations.

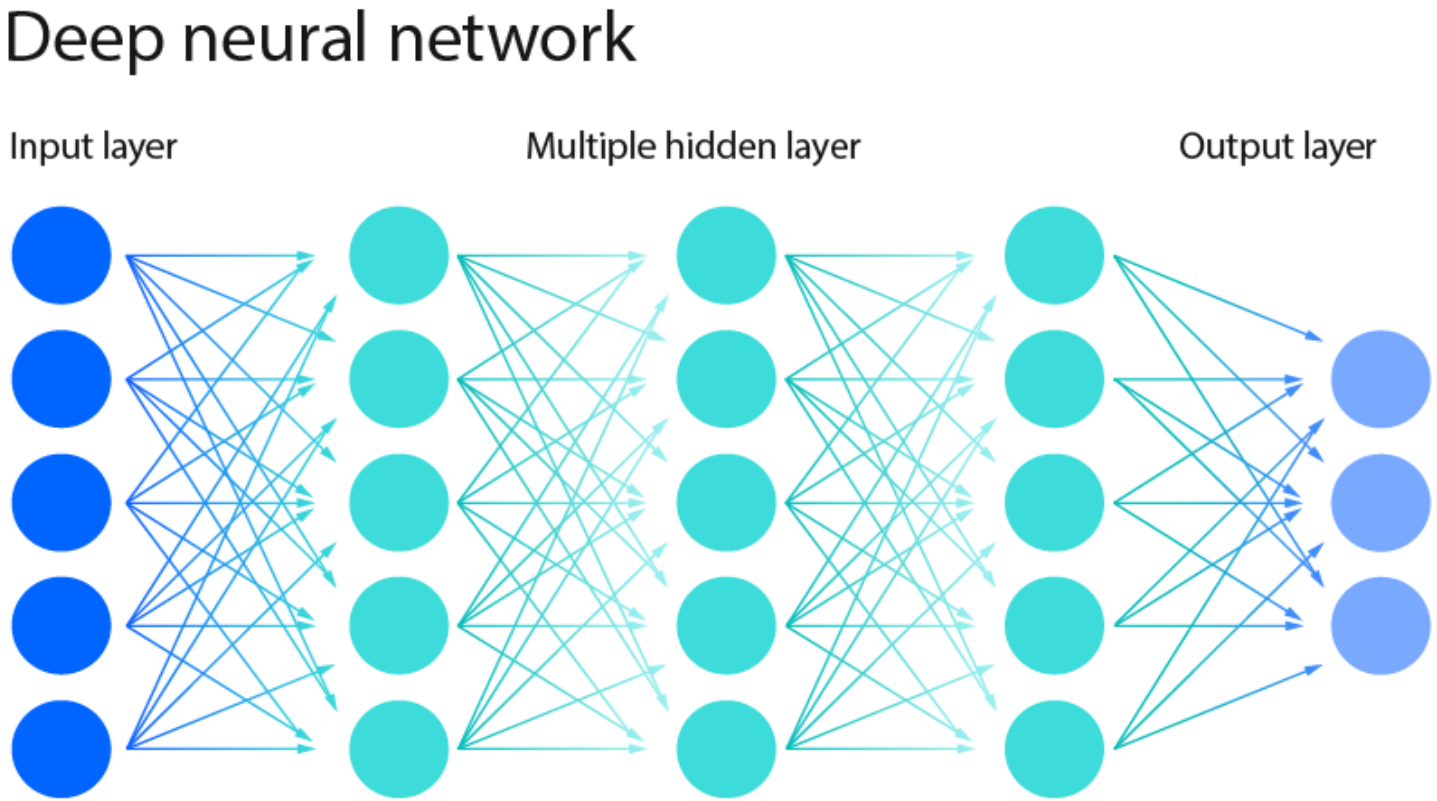
Neuron's Job: Each node receives data, multiplies it by a **weight** (signifying importance), adds a **bias**, and passes the result through an **activation function** to decide if the signal is strong enough to move forward.

Learning: Through a process called **backpropagation**, the network compares its prediction to the correct answer, calculates the error, and adjusts its internal weights to be more accurate next time.



Deep Neural Networks

- A deep neural network is an artificial neural network that contains multiple "hidden layers" between the input and the output.
- We need DNNs because traditional algorithms struggle with **unstructured data** like images, audio, and natural language.
- LLMs are essentially massive DNNs based on the **Transformer** architecture.



Generative AI

Generative AI (Gen AI) is a type of artificial intelligence that can create new, original content – such as text, images, audio, video, and code – rather than just analyzing or classifying existing data.

While traditional AI is designed to act like a "Judge" (e.g., Is this a picture of a cat?), Generative AI acts like an "Artist" or "Writer" (e.g., Draw me a picture of a cat wearing a space suit.).

Gen AI acts as a **Force Multiplier**, shifting the human role from creation from scratch to curation and strategic review.

Large Language Models (LLMs)

An **LLM (Large Language Model)** is a **statistical machine-learning model trained on large amounts of text to predict the next token** (word or sub-word) in a sequence. Everything it can do—reasoning, coding, explanation, conversation—is an *emergent use* of that capability.

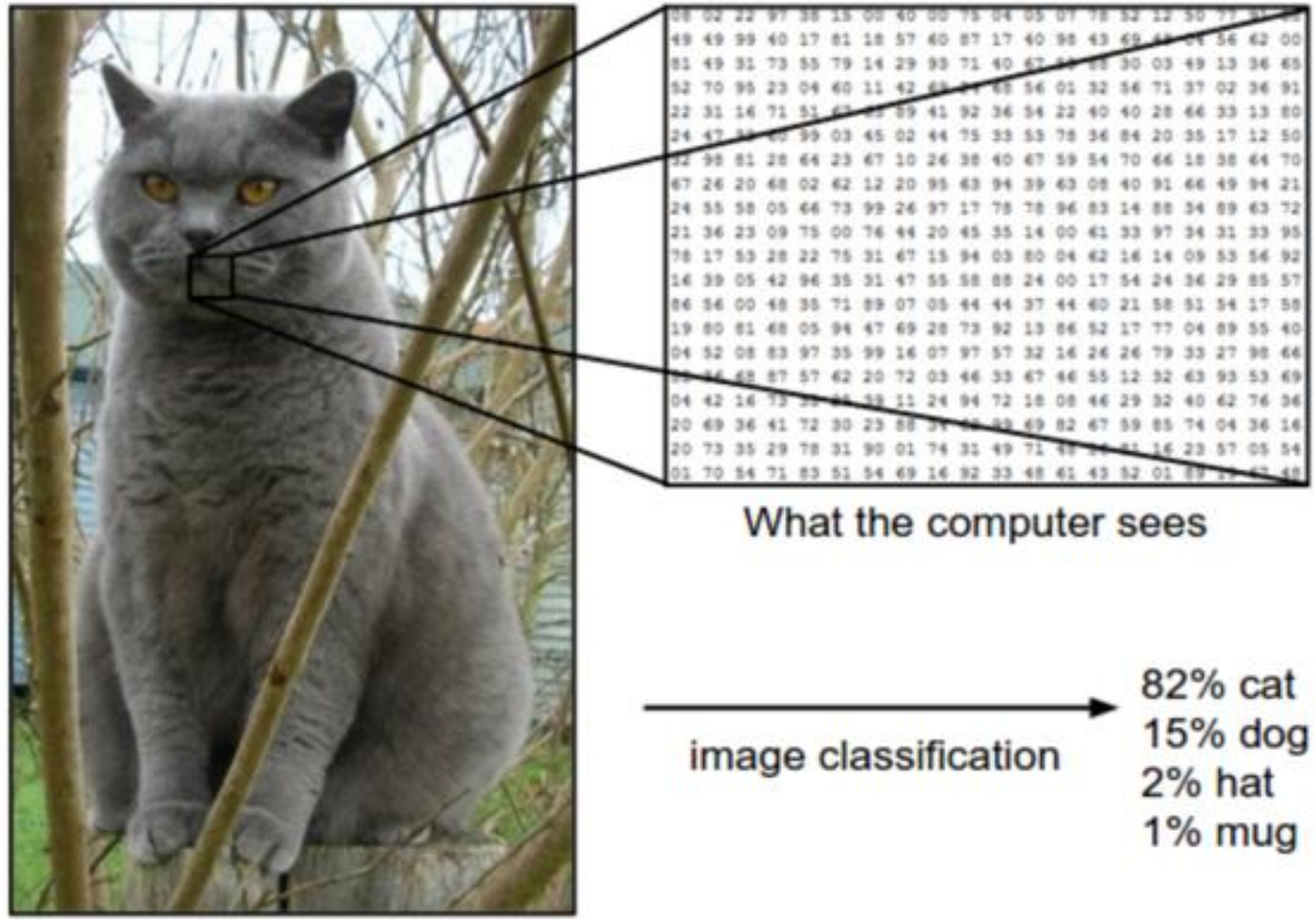
An LLM works through:

1. **Tokenization:** Text is broken down into tokens (not words). Example: "thinking" = ["think", "ing"]
2. **Embeddings:** Each token is converted into a vector (numbers).
   - Similar Tokens: Nearby vectors.
   - This creates a semantic space.
3. **Transformer + Attention:** The core architecture is a transformer. Self-attention answers - "Which previous tokens matter most for predicting the next one?"
4. **Next-token Prediction:** The model computes P(next_token | all_previous_tokens). Sampling or decoding turns probabilities into actual text.

Understanding How AI Thinks

AI systems detect patterns from large data and generate outputs probabilistically — they do not understand, reason, or know truth.

1. AI looks at data (pixels, words, numbers) and finds **correlations**. It learns that "U" often follows "Q" or that a certain cluster of pixels usually means "Cat."
   a. **Truth vs. Pattern:** AI doesn't know what a "Cat" is in the real world. It only knows the **mathematical pattern** that represents a cat in its dataset.
2. When you ask an AI a question, it doesn't "lookup" an answer. It calculates the **likelihood** of the next piece of information.
   a. The Math: If P is the probability, the AI is essentially solving for: $P(\text{word}_n \mid \text{context})$
   b. It chooses the word with the highest statistical weight, not necessarily the one that is "true."
3. AI lacks a "world model." It has no physical experience or moral compass. Because AI is probabilistic, it can "hallucinate." If the most *likely* next word in a pattern is a lie, the AI will output that lie confidently because it matches the statistical pattern of a "correct-sounding" sentence.


What the computer sees
image classification
82% cat
15% dog
2% hat
1% mug

What AI is and what it isn't.

What AI IS (The Function)	What AI IS NOT (The Myth)
Pattern Recognition: The ability to identify regularities in massive, high-dimensional datasets (e.g., spotting a tumor in an X-ray or a fraud pattern in millions of transactions).	**Conscious:** AI has no subjective experience, feelings, or "qualia." It does not "know" it exists; it is a mathematical function executing code.
Statistical Prediction: AI is essentially "high-speed autocomplete." It calculates the mathematical probability of the next pixel, word, or note based on its training data.	**Objective:** AI is a mirror of its training data. Because it learns from human-generated content, it inherits and often amplifies human biases, stereotypes, and cultural prejudices.
Generative Synthesis: Modern models (Transformers/Diffusion) can create "new" content by recombining learned patterns into original arrangements (text, image, video).	**Fact-Aware by Default:** AI does not have a "source of truth." As of 2026, even top models have a **3–15% hallucination rate** on complex tasks because they prioritize "plausibility" over "truth."
Agentic Logic: In 2026, AI can follow multi-step "if-then" reasoning loops to complete tasks, such as booking a flight or writing code.	**Capable of Intent or Judgment:** AI lacks "Agency" in the human sense. It has no personal goals, moral compass, or the ability to understand the consequences of its outputs.

Core Capabilities of GenAI Systems

1. **Content Creation & Drafting**: Generate first-draft emails, articles, and reports in seconds. Condense long documents into concise, actionable executive summaries. High-fidelity language translation and cultural rephrasing for global communication.
2. **Strategic Ideation & Brainstorming**: Generate diverse options for marketing slogans, project names, or product features. Explore multiple "What If" scenarios and structured solution paths. Iterate on initial ideas to improve tone, clarity, or persuasive impact.
3. **Advanced Data & Pattern Recognition**: Identify trends and anomalies in massive volumes of unstructured text or numerical data. Convert raw data into structured insights and key takeaways. Connect disparate ideas across multiple sources to find hidden correlations.
4. **Intelligent Code Assistance:** Produce functional code snippets or entire scripts from natural language prompts. Suggest refactoring options to improve efficiency and maintainability. Assist in debugging and identifying security vulnerabilities (*always requires human review*).

Structural Limitations of GenAI

Generative AI models are probabilistic, not deterministic. This means they predict the "most likely" next word rather than checking a database of facts.

Lack of "Ground Truth" Awareness: Models predict sequences based on patterns, not a built-in understanding of truth or reality. This leads to a behavior called **Hallucinations** in GenAI systems.

Reasoning & Logic Gaps

1. Weak Causal Reasoning: AI is excellent at correlation (finding links) but struggles with causation (understanding why A leads to B).

2. Failure in Novelty: Because they learn from historical data, models struggle to provide accurate insights for brand-new events or truly original problems.

Accountability & Real-World Impact: AI lacks a moral or legal compass; it cannot be held accountable for the advice it gives.

Risks of Generative AI Systems

1. **Privacy & Data Risks**
    a. **Data Leakage:** Users may inadvertently input sensitive, proprietary, or personal data into public AI models, which then becomes part of the training set for future responses.
    b. **Inference Attacks:** AI can analyze "anonymized" data to re-identify individuals or infer sensitive traits (like health conditions or political views) without their consent.
    c. **Surveillance:** AI-powered facial recognition and behavioral tracking can lead to intrusive "social credit" or mass surveillance systems.
2. **Security & Malicious Use**
    a. **Deepfakes & Disinformation:** AI can generate highly realistic fake images, videos, and audio, which can be used for fraud, political manipulation, or destroying reputations.
    b. **Model Poisoning:** Malicious actors can "poison" the data used to train an AI, causing it to have a specific hidden vulnerability or incorrect behavior.
    c. **Prompt Injection:** Tricking an AI (like an LLM) into ignoring its safety guidelines to reveal confidential information or perform unauthorized actions.

Risks in Generative AI Systems

1. **Reliability and "Black Box" Risks**
    a. **Lack of Explainability:** Deep learning models are often "black boxes," meaning even their creators can't explain exactly why the AI made a specific decision. This is high-risk in sectors like healthcare or law.
    b. **Hallucinations:** AI can confidently present false information as a fact. Without "ground truth" awareness, it cannot distinguish between its own inventions and reality.
    c. **Model Drift:** Over time, an AI's performance can degrade as the real-world changes and the data it was originally trained on becomes outdated.
2. **Socioeconomic and Environmental Risks**
    a. **Job Displacement:** Rapid automation of cognitive and creative tasks can lead to significant economic disruption and a widening wealth gap.
    b. **Environmental Impact:** Training "billions of parameters" models requires massive amounts of electricity and water (for cooling data centers), contributing to a large carbon footprint.
    c. **Intellectual Property:** Many models are trained on copyrighted works without the original creators' permission or compensation.

Risks: Hallucination

AI hallucination is when an AI model, especially Generative AI, confidently produces plausible-sounding but factually incorrect, nonsensical, or completely fabricated information, essentially making things up that aren't grounded in its training data or reality, like inventing legal citations or misstating historical facts. This occurs due to complex factors like data biases, model errors, or overfitting, leading to outputs that seem real but are misleading, impacting trust and reliability in applications from chatbots to medical tools.

AI hallucinations are similar to how humans sometimes see figures in the clouds or faces on the moon. In the case of AI, these misinterpretations occur due to various factors, including overfitting, training data bias/inaccuracy and high model complexity.

Other risks include bias from the training data and sycophancy (the tendency of AI systems to agree with the user).

Source: https://cloud.google.com/discover/what-are-ai-hallucinations

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Live Demo: Hallucination

There are various ways you can reproduce the hallucination issue in an LLM.

1. Asking it for today's date.

Strategies to Mitigate AI Risks

To address the structural limitations of Generative AI, organizations use a multi-layered approach that combines technical safeguards with human oversight.

1. **Technical Safeguards (Grounding the Model)**
   - **Retrieval-Augmented Generation (RAG)**: Connects the AI to a trusted, private knowledge base (like your company's documents) to ensure it cites real facts instead of hallucinating.
   - Reinforcement Learning from Human Feedback (RLHF): Continuous training where humans rank AI responses to "tune" the model toward safety, accuracy, and helpfulness.
   - Adversarial Testing (Red Teaming): Rigorous "stress testing" by security teams to find where the model might break or leak data before it is deployed.
2. **Human in the Loop (HITL)**: Establishing a policy where AI-generated content (especially legal, financial, or medical) *must* be reviewed and signed off by a human expert.
3. **Governance Frameworks:** Classifying AI tasks by risk level with stricter approvals for high-risk tiers. Adopting global standards to map, measure, and manage AI-specific vulnerabilities. Automated filters that block the model from seeing or outputting personally identifiable information (PII).

AI Decision Matrix

	Low Complexity/Low Time	High Complexity/High Time
High Stakes	Human-First (Manual Only)	High-Stakes Co-Pilot (Augmentation)
Low Stakes	Experiment Freely (Optional AI)	Start Here (Full AI/Automation)

Responsible AI Use

Before using AI, ask:

1. What is the **task type**? (creative, analytical, judgment-based)
2. What is the **risk of error**?
3. Who is **accountable**?
4. What needs **human review**?

Safe Use Examples

1. First drafts
2. Idea generation
3. Summarization
4. Explaining complex topics

Responsible AI Use

Unsafe / Restricted Use

- Final legal decisions
- Medical or safety-critical advice
- Sensitive HR judgments
- Unverified reporting

Golden Rule

AI accelerates thinking; it does not replace responsibility

Class Quiz

Classify the following tasks for which you can use AI as:

Needs Review/Not Suitable/Safe.

1. Writing a client email.
2. Legal compliance decision.
3. Diagnosing a system failure.

Discuss: AI in the Workplace

In groups of 3-4 people, discuss how you can use AI in your current/previous role:

- One task where AI is appropriate and helpful
- One task where AI should not be used
- One risk you must actively manage when using AI

Takeaways

Responsible AI use is a **professional skill.**

AI is powerful but fundamentally limited.

Confidence is not equal to Correctness.

Risk increases with sensitivity, stakes, and automation.

AI is an Assistant, Not an Authority

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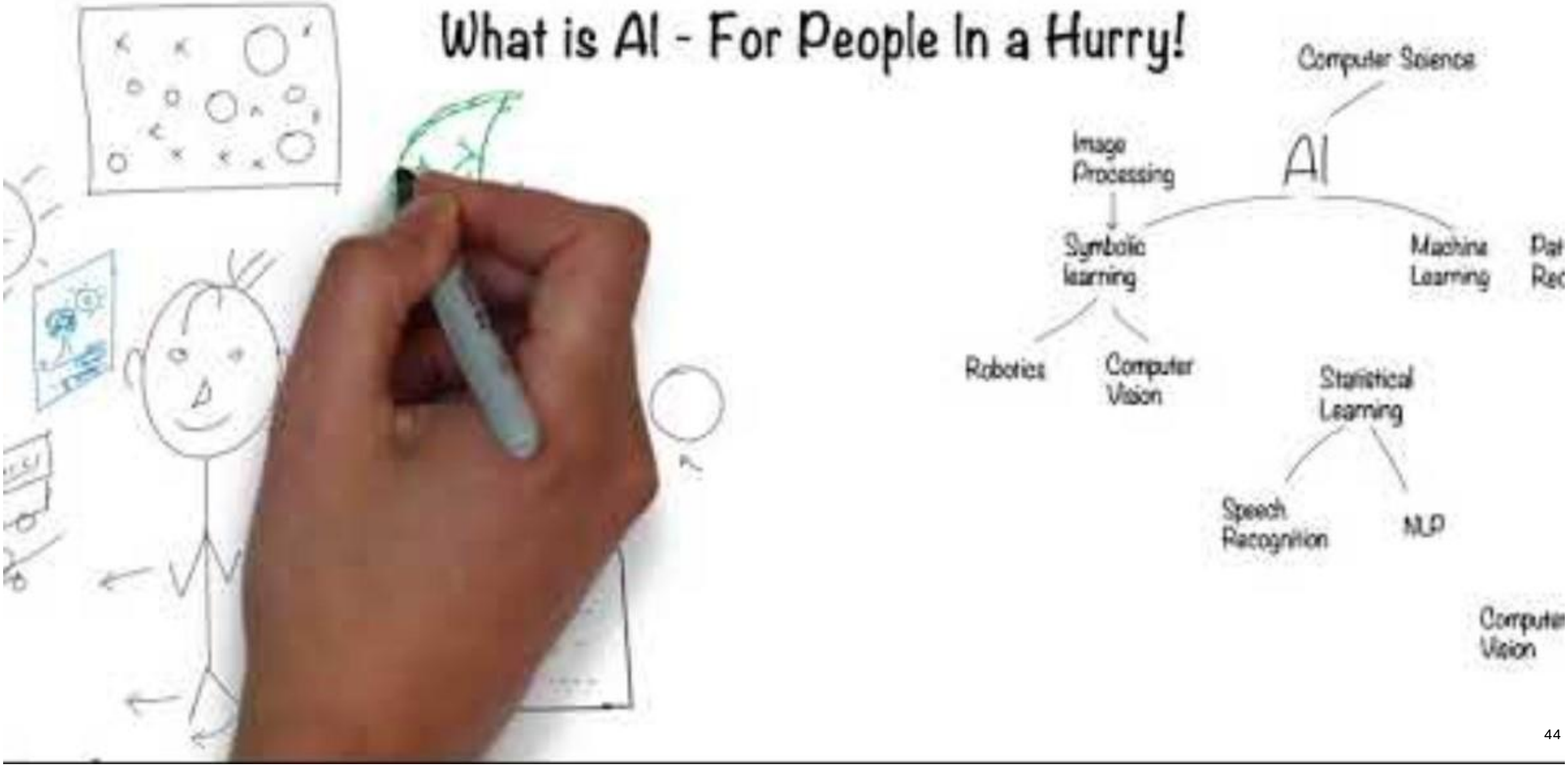
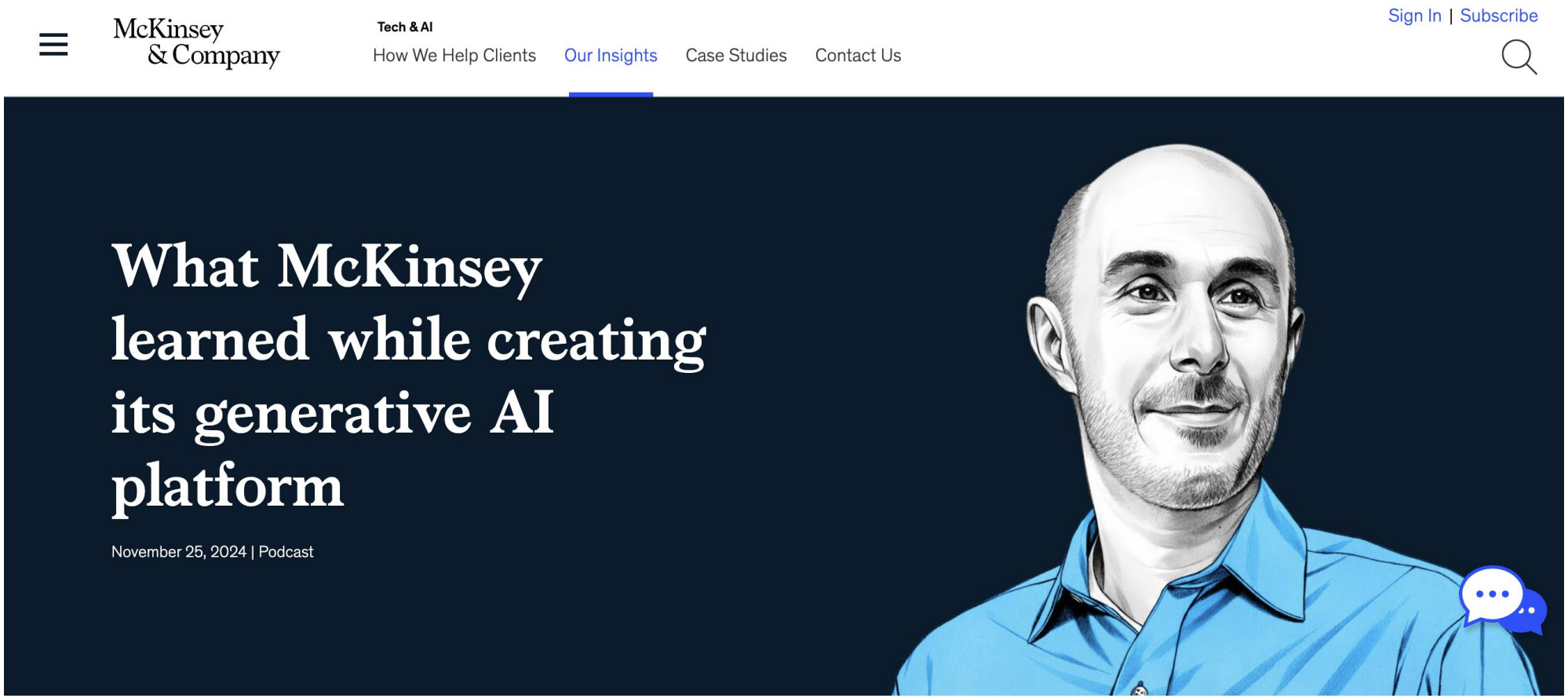

What is AI - For People In a Hurry!
Computer Science
AI
Image Processing
Symbolic learning
Machine Learning
Robotics
Computer Vision
Statistical Learning
Speech Recognition
NLP

Custom GenAI Systems: McKinsey



Source: McKinsey Insights

Thank you!